Light

GCSE Computer Science — Cyber Security Prevention (Set 3)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Cyber Security Prevention

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Software runs untrusted programs in an isolated environment where they cannot access or modify the main system.
What is this security technique called?

- A. Sandboxing
- B. Defragmentation
- C. Data compression
- D. File indexing

Q2 (1 mark)

A computer is configured to only run programs that appear on an approved list; all other software is blocked.
What is this security approach called?

- A. Data mining
- B. Application whitelisting
- C. File compression
- D. Disk formatting

Q3 (1 mark)

A banking website automatically logs users out after 5 minutes of inactivity.
What security benefit does this provide?

- A. It speeds up the website
- B. It reduces internet data usage
- C. It limits the time an unattended session could be misused by someone else
- D. It improves screen brightness

Q4 (1 mark)

An organisation configures its network to block access to known malicious websites by preventing their domain names from resolving.
What security technique is this?

- A. Data compression
- B. File defragmentation
- C. Disk partitioning

- ○ **D.** DNS filtering

Q5 (1 mark)

A computer checks that its startup software has not been tampered with before allowing the operating system to load.
What is this security feature called?

- ○ **A.** Secure boot
- ○ **B.** Fast startup
- ○ **C.** Sleep mode
- ○ **D.** Screen saver

Q6 (1 mark)

An organisation labels documents as 'Public', 'Internal', 'Confidential', or 'Restricted' based on sensitivity.
What is this practice called?

- ○ **A.** File compression
- ○ **B.** Data classification
- ○ **C.** Disk formatting
- ○ **D.** Memory allocation

Q7 (1 mark)

A browser extension blocks advertisements and tracking scripts on websites.
How does this improve security?

- ○ **A.** It makes the computer run faster
- ○ **B.** It increases internet speed
- ○ **C.** It reduces exposure to malvertising attacks where malicious code is delivered through advertisements
- ○ **D.** It improves screen resolution

Q8 (1 mark)

A computer has a guest account with limited permissions that cannot install software or access personal files.
What security principle does this implement?

- ○ **A.** Maximum access for convenience
- ○ **B.** Unlimited sharing permissions
- ○ **C.** Open network access
- ○ **D.** Least privilege by restricting guest capabilities to minimum necessary functions

Q9 (1 mark)

A browser extension automatically upgrades connections from HTTP to HTTPS whenever the secure version is available.
What protection does this provide?

- ○ **A.** Ensures encrypted connections are used whenever possible, protecting data from interception
- ○ **B.** Makes websites load faster
- ○ **C.** Increases available storage space

- ○ **D.** Improves battery life

Q10 (1 mark)

A website asks users to answer personal questions like 'What is your mother's maiden name?' to reset forgotten passwords.
Why are these security questions considered a weak protection?

- ○ **A.** The questions are too difficult to answer
- ○ **B.** Answers can often be found on social media or guessed, making them easy to compromise
- ○ **C.** The questions use too much server storage
- ○ **D.** The questions slow down website loading

Q11 (1 mark)

A user connects to a coffee shop's free WiFi to check their bank account.
What security risk does this create?

- ○ **A.** The coffee is more expensive with WiFi
- ○ **B.** The phone battery drains faster
- ○ **C.** Attackers on the same network could potentially intercept unencrypted traffic or perform man-in-the-middle attacks
- ○ **D.** The phone storage fills up

Q12 (1 mark)

A user only downloads software from official app stores or vendor websites rather than third-party download sites.
Why is this practice recommended?

- ○ **A.** Official sources charge lower prices
- ○ **B.** Third-party sites have slower downloads
- ○ **C.** Unofficial sources always have newer versions
- ○ **D.** Official sources are less likely to contain malware, as apps undergo review and verification processes

Q13 (2 marks)

Software monitors network traffic and endpoint activity to detect and prevent sensitive data (like credit card numbers or personal records) from leaving the organisation without authorisation.
What type of security system is this?

- ○ **A.** Data Loss Prevention (DLP) that identifies and blocks unauthorised transmission of sensitive information
- ○ **B.** Data compression that reduces file sizes
- ○ **C.** Disk defragmentation that improves performance
- ○ **D.** File indexing that speeds up searches

Q14 (2 marks)

Before any modification to IT systems, staff must submit a request describing the change, its purpose, and potential risks. Changes are reviewed and approved before implementation.
What is this process called, and why is it important for security?

- ○ **A.** File compression; it saves storage space

- ○ **B.** Change management; it prevents unauthorised modifications and ensures changes are reviewed for security implications before implementation
- ○ **C.** Data mining; it analyses patterns in information
- ○ **D.** Disk formatting; it prepares storage media

Q15 (2 marks)

A security team subscribes to services that provide information about new attack techniques, emerging malware, and indicators of compromise used by threat actors.
What is this information called, and how does it improve security?

- ○ **A.** File compression data; it reduces storage needs
- ○ **B.** Disk performance metrics; they optimise system speed
- ○ **C.** Threat intelligence; it enables proactive defence by informing teams about threats before they encounter them directly
- ○ **D.** Memory allocation tables; they manage RAM usage

Q16 (2 marks)

A mobile banking app is configured to only accept connections to servers presenting specific, pre-defined security certificates, rejecting even valid certificates from trusted authorities if they do not match.
What is this technique called, and what attack does it prevent?

- ○ **A.** Data compression; it reduces file sizes during transmission
- ○ **B.** Memory caching; it speeds up repeated operations
- ○ **C.** File indexing; it accelerates search functionality

○ **D.** Certificate pinning; it prevents man-in-the-middle attacks using fraudulently obtained certificates

Q17 (2 marks)

An organisation documents standard security settings that must be applied to all new servers, including disabled services, password requirements, and logging configurations.
What is this documented standard called?

○ **A.** A security baseline or hardening guide that defines minimum security configurations

○ **B.** A file compression algorithm

○ **C.** A disk defragmentation schedule

○ **D.** A screen saver configuration

Q18 (2 marks)

A user sets up two-factor authentication but also saves backup codes in a secure location in case they lose access to their authentication device. Why are backup codes important?

○ **A.** Backup codes make the internet faster

○ **B.** Backup codes provide account recovery if the primary authentication device is lost, stolen, or broken — without them, account access could be permanently lost

○ **C.** Backup codes increase storage capacity

○ **D.** Backup codes improve display resolution

Q19 (2 marks)

A building's electronic door locks are designed to remain locked if the power fails, rather than defaulting to unlocked.
What security design principle does this demonstrate?

- ○ **A.** Fail-open design where systems become accessible when they fail
- ○ **B.** Maximum convenience over security
- ○ **C.** Fail-secure design where systems maintain protection even during failures
- ○ **D.** Cost reduction through simpler mechanisms

Q20 (2 marks)

A company configures its systems to block login attempts originating from countries where it has no business operations.
What type of access control is this?

- ○ **A.** File compression based on location
- ○ **B.** Disk defragmentation by region
- ○ **C.** Memory allocation by country
- ○ **D.** Geographic access restrictions that limit connections based on location

Q21 (2 marks)

A security team tracks metrics including: time to detect threats, time to respond to incidents, percentage of systems patched within 30 days, and number of security training completions.
Why is measuring these metrics important?

- ○ **A.** Metrics measure objective security performance, identify improvement areas, demonstrate value to leadership, and enable data-driven decisions
- ○ **B.** Metrics only serve to criticise security staff
- ○ **C.** Metrics replace the need for security controls
- ○ **D.** Metrics are only useful for marketing purposes

Q22 (2 marks)

Before sharing customer data with a third-party service provider, a company assesses the provider's security practices, reviews their certifications, and requires contractual security commitments.
What process is this?

- ○ **A.** File compression assessment
- ○ **B.** Vendor risk management that evaluates third-party security before sharing data
- ○ **C.** Disk defragmentation review
- ○ **D.** Memory optimisation audit

Q23 (2 marks)

A system is configured so that maintenance accounts can only log in during scheduled maintenance windows (Saturday 2am-6am), and are disabled at all other times.
What type of access control enhancement is this?

- ○ **A.** Geographic restrictions based on location
- ○ **B.** Biometric authentication using fingerprints

○ **C.** Time-based access restrictions that limit when accounts can be used

○ **D.** Password complexity requirements

Q24 (3 marks)

An organisation conducts a risk assessment by:

```
• Identifying assets and their value
• Identifying threats that could affect each asset
• Identifying vulnerabilities that threats could exploit
• Estimating likelihood and impact of each risk
• Prioritising risks for treatment
```

Why is formal risk assessment important for security planning?

○ **A.** Risk assessment is unnecessary; all threats should receive equal resources

○ **B.** Risk assessment only matters for compliance paperwork

○ **C.** Risk assessment replaces the need for security controls entirely

○ **D.** Risk assessment enables prioritised resource allocation — limited security budgets should address highest risks first; understanding likelihood and impact ensures efforts focus where they matter most

Q25 (3 marks)

An organisation deploys a Security Information and Event Management (SIEM) system that:
• Collects logs from firewalls, servers, endpoints, and applications

```
• Correlates events across multiple sources
• Applies rules to detect suspicious patterns
```

```
• Generates alerts for security analysts to investigate
```

How does SIEM improve security compared to reviewing individual system logs separately?

- ○ **A.** SIEM correlates events across systems to detect attacks that span multiple components — an attacker's activities might appear normal on individual systems but reveal patterns when viewed together; centralised collection also prevents attackers from hiding their tracks by deleting local logs
- ○ **B.** SIEM only stores logs without any analysis capabilities
- ○ **C.** Individual log review is always more effective than centralised systems
- ○ **D.** SIEM replaces the need for security controls by detecting all attacks automatically

Q26 (3 marks)

A company transitions from traditional perimeter security to Zero Trust architecture by:

```
• Requiring authentication for every access request, not just at
• Verifying device health before granting access
• Applying micro-segmentation so systems can only communicate wit
• Continuously monitoring and logging all access
```

Why is Zero Trust increasingly adopted over traditional perimeter security?

- ○ **A.** Traditional perimeter security is always more effective than Zero Trust
- ○ **B.** Zero Trust addresses modern realities: users work remotely, applications run in cloud, devices connect from anywhere, and threats exist inside networks — traditional perimeter assumptions that 'inside is trusted' no longer hold true

○ **C.** Zero Trust eliminates the need for any access controls

○ **D.** Zero Trust only works for small organisations

Q27 (3 marks)

A security programme assessment finds:
• Technical controls are well-implemented
• Policies are comprehensive and documented
• However, employees routinely share passwords, hold doors for strangers, and click suspicious links despite training
What does this reveal about the organisation's security posture, and how should it be addressed?

○ **A.** Technical controls are all that matter; human behaviour is irrelevant

○ **B.** More policies will solve the behavioural problems

○ **C.** Security culture is weak despite good technical and policy foundations; addressing this requires ongoing engagement, leadership example, making security convenient, positive reinforcement, and integrating security into daily workflows rather than treating it as an obstacle

○ **D.** Removing all technical controls will improve human behaviour

Q28 (3 marks)

A security analyst compares traditional antivirus with Endpoint Detection and Response (EDR):
Traditional antivirus: Signature-based detection, blocks known threats, limited visibility
EDR: Behavioural analysis, continuous monitoring, threat hunting, incident investigation, response capabilities
Why have many organisations deployed EDR alongside or replacing traditional antivirus?

○ **A.** Traditional antivirus catches all possible threats; EDR provides no additional value

○ **B.** EDR is only useful for very small organisations

○ **C.** EDR only performs the same functions as antivirus

○ **D.** EDR detects threats that evade signatures through behaviour analysis, provides visibility into endpoint activities for threat hunting, enables investigation of incidents to understand attack scope, and offers response capabilities to contain threats — addressing limitations of signature-only detection

Q29 (3 marks)

A security architect designs a new application following these principles:
• Minimise attack surface by disabling unused features
• Apply defence in depth with multiple control layers
• Fail securely with safe defaults during errors
• Use least privilege for all components
• Separate duties for sensitive operations
Why should security be designed into systems from the start rather than added later?

○ **A.** Security designed from the start is integrated into architecture at lower cost; retrofitting security onto completed systems is expensive, often incomplete, and may require fundamental redesign; early security decisions shape what is possible later

○ **B.** Security can always be added easily at any stage without additional cost

○ **C.** Design-time security provides no advantages over later additions

○ **D.** Security should only be considered after all functionality is complete

Q30 (3 marks)

An organisation establishes a Security Operations Centre (SOC) that:
- Monitors security systems 24/7
- Triages and investigates alerts
- Coordinates incident response
- Performs threat hunting
- Produces security reports for management

What value does a dedicated SOC provide compared to security being a part-time responsibility distributed among IT staff?

- ○ **A.** Distributed part-time security is always more effective than dedicated teams
- ○ **B.** A dedicated SOC provides focused expertise, continuous monitoring, faster response, consistent processes, and accumulated institutional knowledge — security threats do not respect business hours and require specialised skills that part-time attention cannot develop
- ○ **C.** SOCs only generate reports without operational value
- ○ **D.** Security monitoring requires no specialised skills or dedicated attention

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.